HOJUN · CLASS 13

Jackets, Not Piles

Four jacket sites. Read them, say them back, then talk about Wando.

SITE 1 · SCOTLAND

Seagreen

THE SITE · 읽고 이해하기

· SSE Renewables + TotalEnergies · fully operational 2023 · 1,075 MW
· 114 Vestas V164 turbines, 10 MW each · 27 km offshore
· Water depth 42–58.6 m — the world's deepest bottom-fixed foundation
· Three-legged jackets with suction buckets, about 2,000 tonnes each
· Installed by the Saipem 7000 semi-submersible crane vessel

READ What are the three or four numbers that matter here?

이 현장에서 중요한 숫자 서너 개만 짚어보기

SAY IT BACK Tell me what they built, in your own words.

읽은 걸 본인 말로 다시 설명하기

COMPARE How is this different from Wando?

완도와 뭐가 다른지

TERMS · 용어

bottom-fixed 고정식 기초
suction bucket 석션 버킷
three-legged jacket 3 각 자켓
water depth 수심
crane vessel 크레인선
semi-submersible 반잠수식
tonnes 톤

SAYING IT · 말할 때

It's a ~ project in ~ ~ 에 있는 ~ 프로젝트예요
They used ~ instead of ~ ~ 대신 ~ 를 썼어요
The water there is ~ deep 거기 수심은 ~ 예요
Ours is ~, so ~ 저희는 ~ 라서 ~ 예요
I'd have to check that 그건 확인해봐야 해요

FROM YOUR WORDS · 지난주

submit
solid

SITE 2 · TAIWAN

Greater Changhua 1 & 2a

THE SITE · 읽고 이해하기

· Ørsted · commissioned 2023 · 900 MW · 35–50 km offshore
· 111 Siemens Gamesa 8 MW turbines on three-legged jackets
· Jackets over 1,200 tonnes, 60–80 m tall
· 333 pin piles — about 4 m across, 80–90 m long, 400 tonnes each
· Designed for typhoons and earthquakes; turbines shut down above 25 m/s
· Over 60% of the pin piles were made by Taiwanese suppliers

READ Pin piles, not a monopile. What's the difference?

핀파일과 모노파일의 차이를 말로

SAY IT BACK Explain the foundation to someone who's never seen one.

기초를 처음 보는 사람에게 설명하듯

COMPARE Wando has typhoons too. What does that change?

완도도 태풍이 있죠 — 그래서 뭐가 달라지나

TERMS · 용어

pin pile 핀파일
driven pile 항타 말뚝
transition piece 전이부
typhoon design 태풍 설계
seismic 지진의
local supplier 현지 공급사
fabrication 제작

SAYING IT · 말할 때

Each pile is about ~ 말뚝 하나가 대략 ~ 예요
They drive them into ~ ~ 에 박아 넣어요
It has to stand up to ~ ~ 를 견뎌야 해요
We do the same, except ~ 저희도 같은데 , 다만 ~
That's outside what I've worked on 그건 제가 안 해본 부분이에요

FROM YOUR WORDS · 지난주

restrict　tight

SITE 3 · GERMANY

Borkum Riffgrund 2

THE SITE · 읽고 이해하기

· Ørsted · 2019 · 450 MW · 34–54 km offshore · water depth 27–33 m
· 56 turbines at 8 MW
· Two foundation types in one farm: 20 suction bucket jackets + 36 monopiles
· Each suction bucket jacket is over 50 m tall and weighs 950 tonnes
· No pile driving — so no underwater hammering noise

READ Why would one wind farm use two foundation types?

한 단지에 기초 두 종류를 쓴 이유는

SAY IT BACK What does a suction bucket actually do?

석션 버킷이 무슨 일을 하는지

COMPARE Could this work at Wando? Why, or why not?

완도에서도 될까 , 안 될까

TERMS · 용어

suction bucket 석션 버킷
pile driving 항타
seabed conditions 해저 지반 조건
soil type 토질
noise mitigation 소음 저감
bearing capacity 지지력

SAYING IT · 말할 때

It depends on the seabed 해저 지반에 따라 달라요
The advantage is ~ 장점은 ~ 예요
The risk is ~ 위험한 건 ~ 예요
I'm not sure — I'd guess ~ 확실치 않은데 , 아마 ~
We'd never do that because ~ 저흰 ~ 때문에 안 해요

FROM YOUR WORDS · 지난주

unlikely
assure

SITE 4 · TAIWAN

Hai Long

THE SITE · 읽고 이해하기

· Commissioned 2025 · 73 Siemens Gamesa 14-222 DD turbines
· 45–70 km off the Changhua coast, Taiwan Strait
· All 73 jacket foundations on pin piles, installed by CDWE
· Record pace: three jackets installed in two days
· All foundations finished ahead of schedule, using the vessel Green Jade

READ Three jackets in two days. What has to go right for that?

이틀에 세 기 — 뭐가 맞아떨어져야 하나

SAY IT BACK Tell me how a jacket gets from the yard to the seabed.

야드에서 해저까지 자켓이 가는 과정

COMPARE What would you want to ask that team?

그 팀에 뭘 물어보고 싶은지

TERMS · 용어

load-out 선적
weather window 작업 가능 기상 조건
installation vessel 설치선
ahead of schedule 일정보다 앞서
campaign 설치 작업 기간
mobilisation 장비 투입

SAYING IT · 말할 때

First they ~, then they ~ 먼저 ~하고, 그다음 ~해요
It takes about ~ per unit 한 기에 ~ 정도 걸려요
The weather decides it 날씨가 결정해요
I'd want to ask them about ~ ~에 대해 물어보고 싶어요
We're slower because ~ 저흰 ~ 때문에 더 느려요

FROM YOUR WORDS · 지난주

preparation　arrival

YOUR SITE

Now Tell Me About Wando

QUESTIONS

Q1 What's the water depth at your site?

Q2 How many legs, how many piles?

Q3 Who's fabricating your jackets?

Q4 What's the schedule this year?

Q5 What's the part you're least sure about?

TOOLBOX

At Wando, the depth is ~
완도 수심은 ~예요

We're using ~ legs and ~ piles
다리 ~개, 말뚝 ~개예요

~ is fabricating ours
저희 건 ~에서 제작해요

We're planning to ~ by ~
~까지 ~할 계획이에요

What I'm least sure about is ~
제일 확신이 안 서는 건 ~예요

I'd have to check the drawings
도면을 봐야 알겠어요

FROM YOUR WORDS · 지난주 단어

specialist

framework

CAUGHT TODAY